MA 301
Test 1
Spring - 2014

Problem	points	out of
1		5
2		5
3		5
4		5
5		10
6		8
Score		38

Name: ____________________

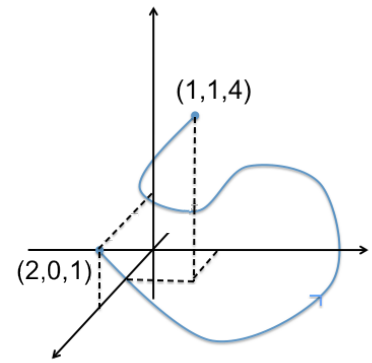
HR: ____________________

Signature: ____________________

READ BEFORE YOU START
* Show all work.
* Change your calculator mode to RADIANS.
* Sloppy solutions may be penalized.
* Use calculators ONLY for computations.
* You must write up every problem in a clear and organized way.
* If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Let **F**$= [z^2, x-y, 7]$ and C be the intersection curve of the surfaces $x = y + z$ and $y = z^2$. For $0 \leq y \leq 9$, compute $\int_C \mathbf{F} \cdot d\mathbf{r}$

Problem 2: Let $\mathbf{F} = \left[\dfrac{y^2}{x}, 2y\ln x, 3z^2\right]$ compute the work required to move an object along C (shown below). Justify your work.



Problem 3: Let C be the triangle with vertices $(0,0)$, $(1,0)$, $(1,4)$ oriented counter-clockwise and

$$\mathbf{F}(x,y) = (xe^y - 3xy)\mathbf{i} + \frac{3}{2}y^2\mathbf{j}$$

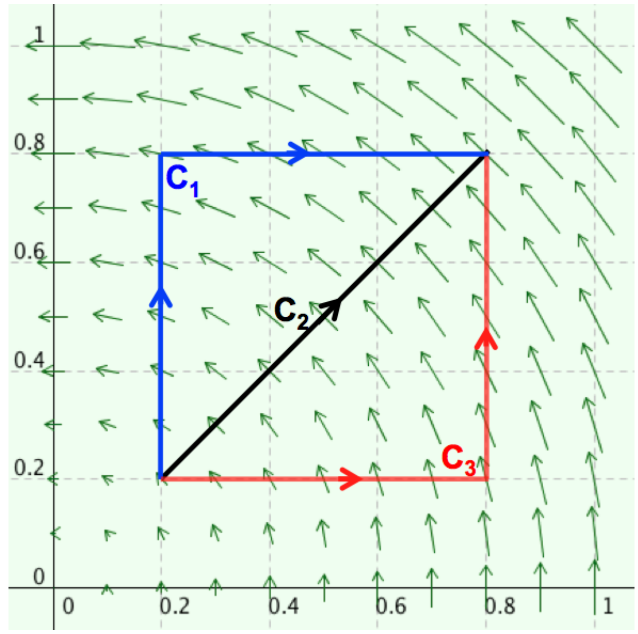
Find the flux of $\mathbf{F}$ around C: $\int_C \mathbf{F} \cdot \mathbf{N}\, ds$ where $\mathbf{N}$ is the unit normal to C

Problem 4: Let R be a region bounded by the upper half of the circle $x^2 + y^2 = 10$ and the x-axis. Let C be the boundary of R with counter-clockwise orientation. Compute the following line integral

$$\int_C (15x - 4xy)\,dx + (\cos y - 6xy)\,dy$$

Problem 5: Answer the following questions.

(a) Consider the vector field $\mathbf{F}$ shown in the figure, together with the paths C_1, C_2, and C_3. Order the line integrals $\left(\int_{C_1} \mathbf{F} \cdot d\mathbf{r},\ \int_{C_2} \mathbf{F} \cdot d\mathbf{r},\ \int_{C_3} \mathbf{F} \cdot d\mathbf{r}\right)$ from smallest to largest. Explain your reasoning.



(b) Suppose $\mathbf{F}$ is conservative and $\int_{(0,0)}^{(0,1)} \mathbf{F} \cdot d\mathbf{r} = 2.4$, $\int_{(0,1)}^{(1,1)} \mathbf{F} \cdot d\mathbf{r} = -1.2$ and $\int_{(0,0)}^{(1,0)} \mathbf{F} \cdot d\mathbf{r} = 7.5$. Then

$$\int_{(1,0)}^{(1,1)} \mathbf{F} \cdot d\mathbf{r} =$$

(c) Let $\mathbf{F} = [-y, x]$ and C be a simple closed curve in the $xy-$plane.

Show that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2 \cdot \text{Area}(R)$ where R is the region bounded by C.

Problem 6: Let $\mathbf{F} = [x + y, 2xy]$ and C be the boundary of the region enclosed by the curves $y = x^2$, $y = x + 2$, and $x = 0$ in the first quadrant.

(a) Setup the integral you would use to compute $\oint\limits_C \mathbf{F} \cdot d\mathbf{r}$ directly (*don't leave integrand as a dot product*)

(b) Compute $\oint\limits_C \mathbf{F} \cdot d\mathbf{r}$ using green's theorem.